

Confidence Augmented Bootstrap Percolation

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The Setup

The existing baseline (Janson) gives a bank exactly **one way to fail**: enough of its neighbors have already activated.

Janson et al. solved that case on the Erdős–Rényi graph, as bootstrap percolation with threshold r . I add a **second way for nodes to activate**.

The Model

A node is activated if either condition is met:

1. **Percolation**: At least r of its neighbors have failed
($|\{j \sim i : j \in A_t\}| \geq r$).
2. **Fear**: A Bernoulli variable with probability $f_i g_t$ that a node activates ($U_i < f_i g_t$).

The Comparisons

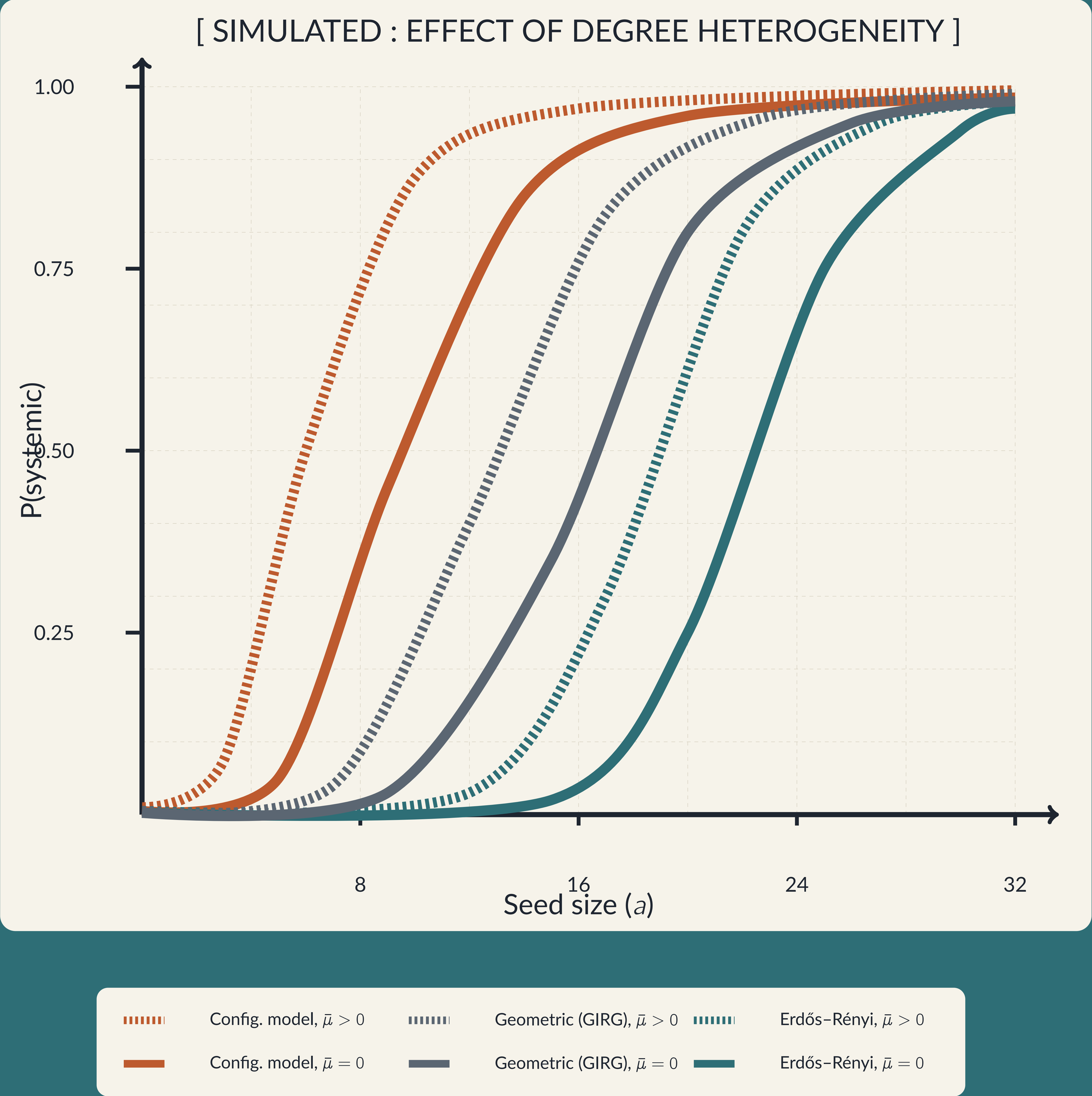
We hold network size n and average degree constant across all tests to isolate three items:

- **Heterogeneity (ER vs. CM)**: Homogeneous degree distribution vs. heavy-tailed hubs.
- **Geometry (CM vs. GIRG)**: Pure random connections vs. local spatial clustering.
- **Fear ($\bar{\mu} = 0$ vs. $\bar{\mu} > 0$)**: Baseline bootstrap percolation vs. confidence augmentation.



Scan to interact with live graphs

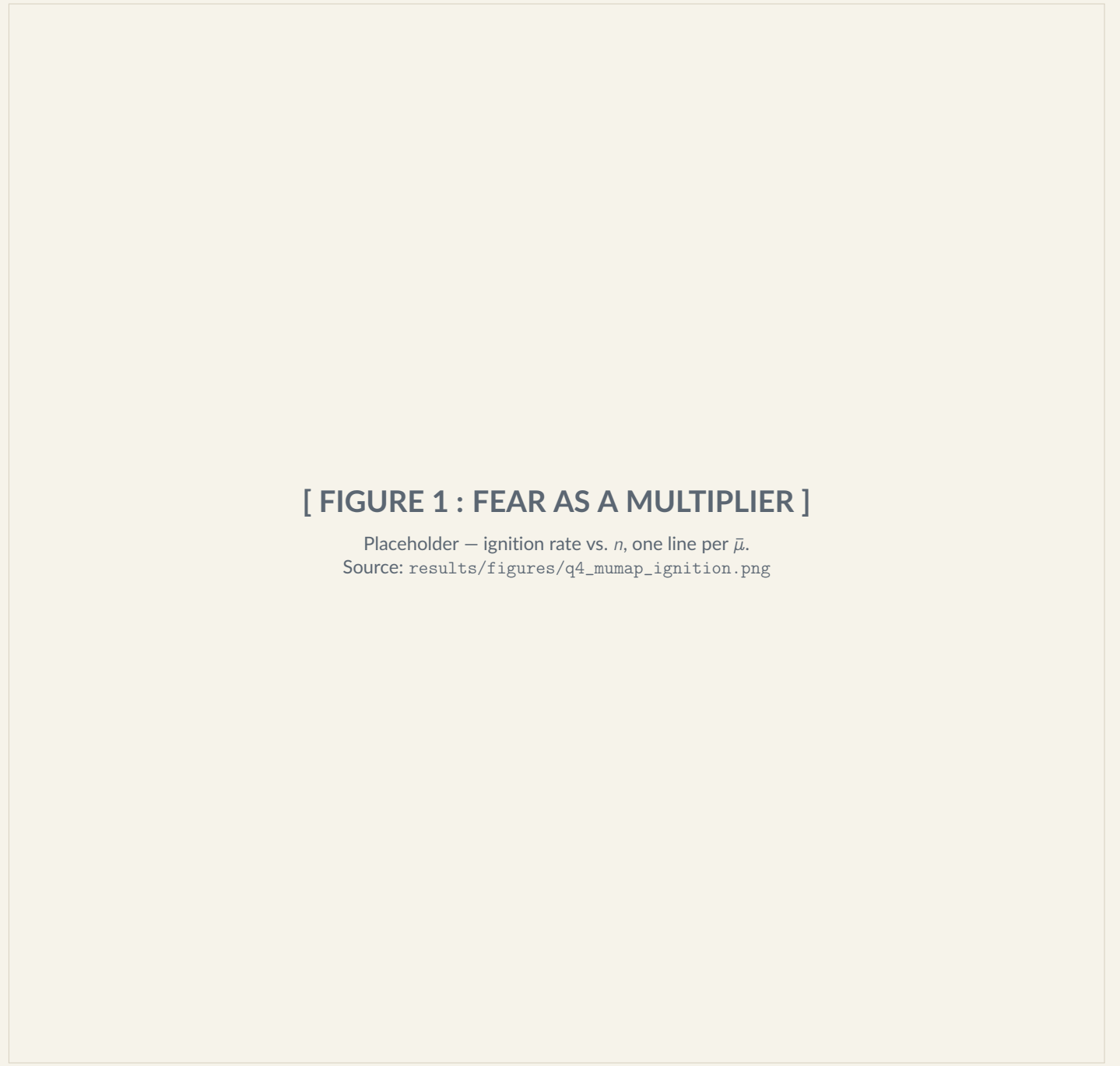
Who you are connected to decides whether a panic ignites. Fear decides how far it gets.



Simulated placeholder for layout evaluation only. Matched n and mean degree;

Fear Amplifies

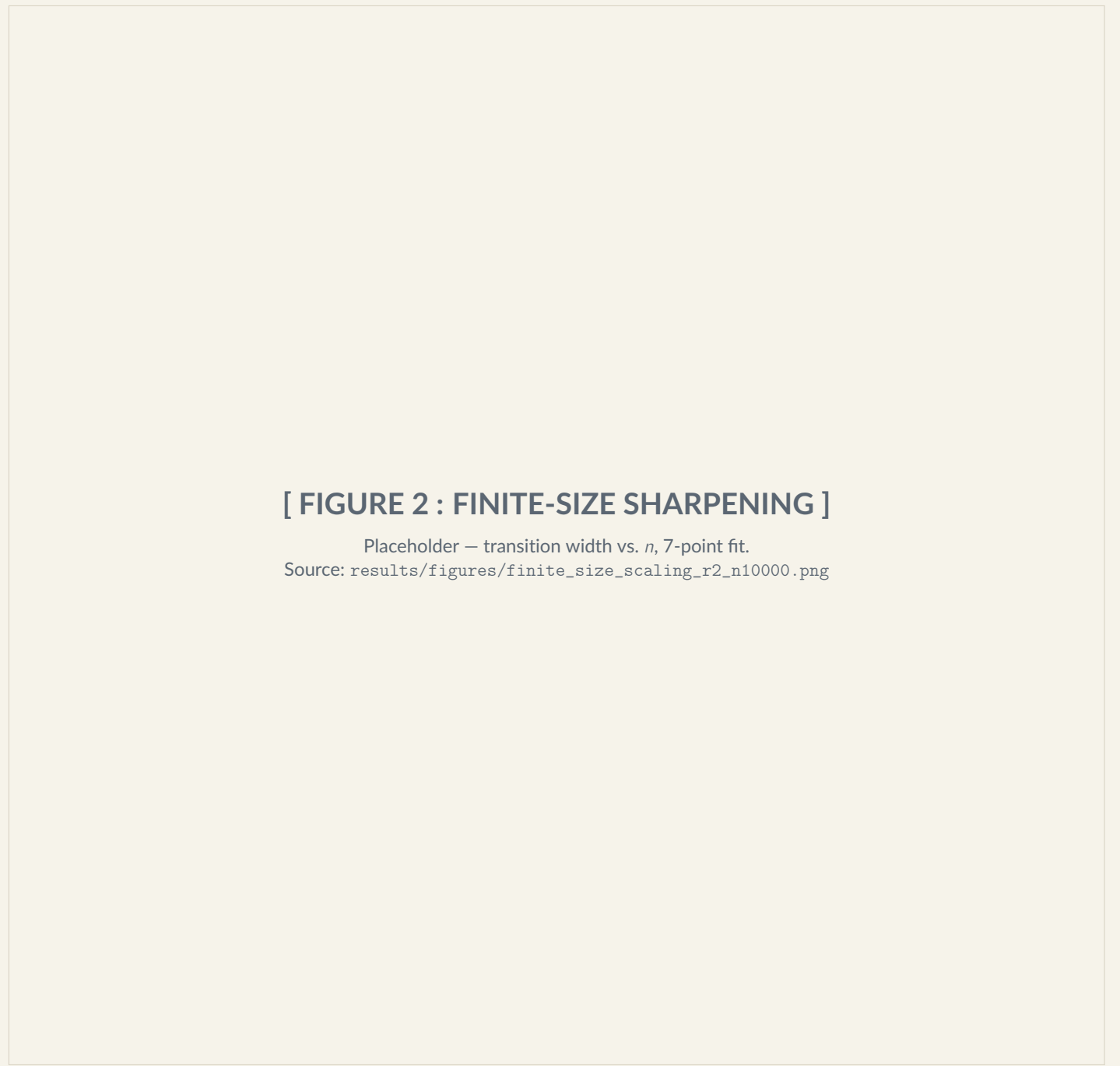
Fear multiplies the ignition probability by roughly $3\times$, and that multiplier is **flat** in n at fixed $\bar{\mu}$, fear changes the *level* of ignition, not its **scaling**. ($\bar{\mu} = 0.1$), so it is an amplifier with a dial, not a constant.



Sharper Transition

Fear makes the systemic boundary **sharper**, leaving the system less forgiving of small perturbations.

$$\nu(\mu = 0) = 5.61 \pm 0.18$$
$$\nu(\mu = 0.3) = 4.82 \pm 0.15$$



Where To Intervene

- **Calm the hubs**: Tilting fear reduction toward high-degree nodes lowers the critical seed monotonically.
- **Limit visibility**: When the fear field is **local** rather than global, cascades behave like the zero-fear baseline. Panic requires a global audience.

Next Steps

We are applying this framework to real-world SNAP networks.